

nose of the Gemini capsule then fit into a docking collar on the Agena.

To avoid long delays between flights, Gemini spacecraft was made more serviceable, with subsystems that could be removed and replaced easily. An adapter module fitted to the rear of the capsule (and jettisoned before re-entry) carried on-board oxygen, fuel and other consumable supplies.

Gemini gave US astronauts their first real experience with living and working in space. They had to learn to sleep and keep house on long flights in crowded quarters, both of which were difficult. These astronauts also made the first forays outside their spacecraft, which required a new spacesuit design. Space walks, however, proved more difficult than expected.

The Gemini IV mission – the second manned flight in the programme – saw Ed White become the first American to perform extravehicular activity (EVA). On June 3, 1965, White stepped out of the spacecraft and used a zip-gun (which propels the astronaut using pressurised oxygen) to go out as far as 15 feet from the capsule. So enjoyable was the experience that White refused to come back in even when ordered by Command Pilot James McDivitt. White tried making up excuses of having to take photographs before finally succumbing to the orders. As he was re-entering the spacecraft, he sighed, “This is the saddest moment of my life.”

It was during the Gemini program that space flight became routine. Ten manned missions left the launch pads of Cape Canaveral in less than 20 months, and the Manned Spacecraft Centre outside Houston (later Johnson Space Centre) took over the role of Mission Control. Ground operations became smooth and efficient, due in part to fleetingly short launch windows – the Gemini XI “window” opened for only two seconds – dictated by the need to rendezvous with targets already in orbit. Meanwhile, sixteen new astronauts chalked up experience in space.

By Gemini’s end, an important new capability – orbital rendezvous and



Ed White becomes the first American to go on a spacewalk